Leveraging Pretrained Language Models for Real-Time Sentiment Analysis in Transliterated Bengali Text

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It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

A new approach to sentiment analysis is studied in this thesis regarding to the Bengali language, which is written using English alphabets. For sentiment analysis, traditional methods such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Natural Language Processing (NLP) algorithms have been extensively implemented. On the contrary, the aim of this research is to apply transformer-based Large Language Models (LLMs) that have undergone pretrained training, in order to propose an alternative methodology for predicting sentiment in real-time.The primary goal of this research initiative is to improve and optimize existing LLMs with the purpose of accurately determining the feelings expressed in Bengali sentences that have been written in English . We are going to use a variety of transliterated Bengali text to teach the LLM how to understand the subtleties and difficulties of expressing emotion in this language setting.Modern language models are used by the suggested real-time sentiment analysis engine to actively guess how people feel about Bengali phrases written with English characters. As part of the study, transfer learning methods, model design improvements, and speed optimization techniques are all being looked into in depth in order to make the LLM better at predicting sentiment.The study's results could be used in lots of different areas like social media monitoring, analyzing of customers feedback, and predicting sentiment about the market.This study improves the field of natural language processing by expanding the use of LLMs to tackle language-specific obstacles in sentiment analysis, specifically in languages that utilize non-Latin letters. This research deploys the Llama2-7B as a pre-trained LLM model.However our proposed model shows better performance with an accuracy of 83.47% and training time of 40.02 minutes .

Keywords: Transliterated Bengali Text, Large Language Models (LLMs), Pre-trained Models, Natural Language Processing (NLP), Emotion Expression, Real-Time Prediction.

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Chapter 1

Introduction

A new area of computer linguistics called "sentiment analysis" looks closely at opinions, feelings, and attitudes that are expressed in written language. Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Natural Language Processing (NLP) algorithms are some of the most important traditional methods used in this field. SVM tries to figure out how people feel by finding the best lines that divide the different types of feelings in the feature space. It is known for being good at classifying things. KNN, on the other hand, uses closeness in the feature space to classify emotion by looking at the feelings of cases that are close by. NLP systems that understand language break down text for clues about how people feel by analyzing its syntax and meaning.
This study proposed the optimized engine which can generate the multi-classification sentiment(positive,negative and neutral) of transliterated Bengali text. Transliterated Bengali text refers to Bengali language text that has been converted from its native script(Bengali script) into the English alphabet.In the case of training our engine ,we used a pre-trained Large Language Models (LLMs) which is come up with a novel approach about fine-tuning our model for generating the sentiment analysis of transliterated Bengali text.However,our proposed LLM model have already have been trained on billions amount of language data.For our research purpose,by fine-tuning this pre-trained LLM model can contribute an important role to understand about the complex contextual meaning,sarcasm and nuances of Transliterated Bengali text.Moreover,this study also explain the benefits and limitations of both traditional methods and advanced Language Learning Models (LLMs) via a comparative analysis.This paper adds to the current discussion on mood analysis by showing how several methods can be used

1.1 Research Background

The research gap in the existing literature is evident when it comes to sentiment analysis in Bengali text transliterated into English.Existing methods are insufficiently advanced to accurately capture all of the nuances and emotions expressed in this kind of communication. The possible consequences of overcoming this gap via LLMs are significant.These models, such as GPT-3.5, have demonstrated superior performance in various natural language processing tasks, and adapting them to handle transliterated Bengali text could significantly enhance sentiment analysis accuracy.The objective of the project is to enhance and modify LLMs in order to

accurately understand the emotions expressed in transliterated Bengali script. The main objective is to create a real-time mood analysis system that can read English-written Bengali words. This objective reflects the rising need for emotions analysis tools that work well on modern communication and social media sites where prompt responses are important for giving opinions.
The study's scope is limited by the linguistic and contextual limitations of Bengali text written in English Alphabets . Finding solutions to the problems that arise up with this way of representing language is the primary objective of the work..The suggested sentiment analysis tool can be used in many ways, such as to track social media, look at customer comments, and guess how people will feel about the market.

1.2 Motivation

LLMs model acts best in the case of cost-effectiveness and potential for real-time sentiment analysis about the contextual meaning of transliterated texts especially bengali text written in English alphabet.That's why this study shows interest to use the pretrained LLMs model.Meanwhile,Traditional methods such as SVM,KNN or NLP face limitations for real-time sentiment analyze .For example, it takes long duration for training the datasets .Besides,it also requires the huge computational resources.In addition ,for real-time sentiment analysis,there is big chance to fail to generate the accurate sentiment of complex contextual meaning. On the other hand ,LLMs model are more advanced because LLMs models are pre-trained on millions or of textual data.Moreover LLMs models work with parameter sharing across layers that means the knowledge learned in one layer is shared across the entire model.That's why in the case of fine-tuning for specific tasks, this pretrained LLMs model is very effective for pattern learning and also reduce the training time.For this advantages of LLMs,it can be more optimized in the case of real-time sentiment of transliterated bengali texts.This study discusses about the implementation of LLMs engine for generating sentiment analysis for transliterated Bengali text and also shows the benefits of using the pre-trained LLM's model in spite of using traditional machine learning method.

1.3 Problem Statement

There are several limitations in the case of fine-tuning the pre-trained large language models (LLMs) for sentiment analysis tasks.One of the most important constraints is that pre-trained LLMs models demand high usage of Video Random Access Memory(VRAM) . LLMs model is pre-trained with millions or even billions of amount of parameters.That's why it's takes high VRAM usage in the case of fine-tuning for sentiment analysis task.After deploying this pre-trained model for our proposed fine-tuning engine ,it faces several problems .For example,for the devices with limited graphical processing capabilities can lead to slower computations as well as increase latency for real-time sentiment analysis applications.However,to overcome this problem,this research shows the effective solution by using the functionality of fully sharded data parallel(FSDF) which is proposed by Zhao, Y. et al. [1]. Actually the FSDP method breaks down the model instance into smaller components and processes each component separately.By applying this idea in this research,memory

efficiency is enhanced by sharding model parameters, gradients, and optimizer states over many GPUs.Moreover,computational efficiency is improved by breaking down communication and integrating it with both the forward and backward passes. Con-

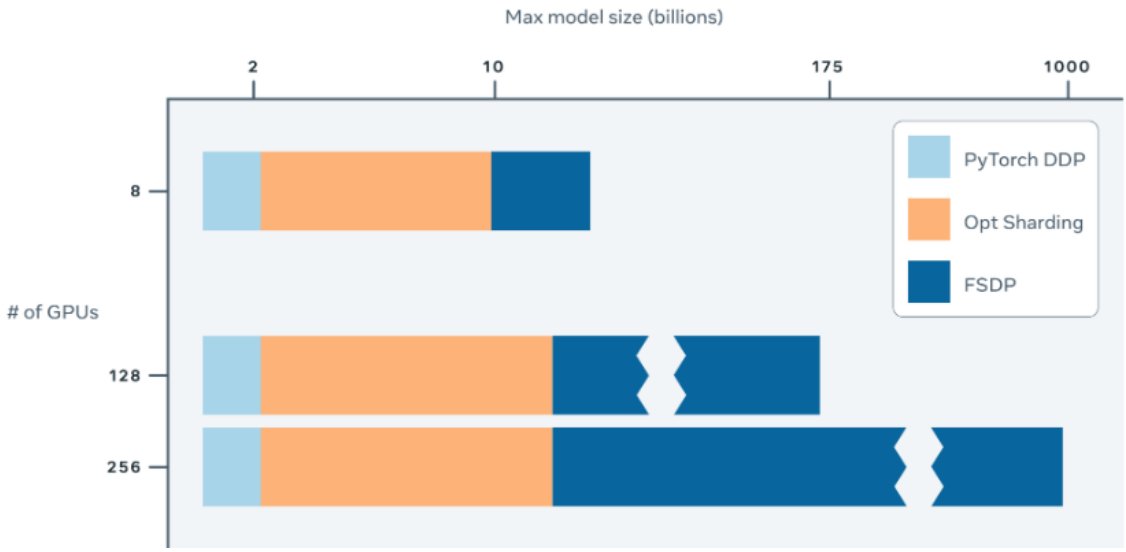


Figure 1.1: Overview of FSDP

text window shifting is another significant constraint. This limitation occurs in the case of smaller context windows because it causes memory issues.That's why LLMs model might have a huge chance to forget some necessary details mentioned earlier.But in the case of larger window allows it to remember.Moreover the speed and cost of running the model is also depend on the size of the context window.As our text datasets for sentiment analysis involve generating long-form content,smaller context window is one of the primary challenge .As our goal is to Deploy a real-time sentiment analysis engine, device with low graphics processing might face difficulties.Additionally, our primary aim is to understand the contextual meaning of text,shifting into smaller context window may impact the speed and the cost of running the model.However to overcome this solution ,this research implemented the functionality of Attention sink Mechanism-streaming Algorithm which is proposed by Xiao, G. et al. [2]. By using this mechanism ,this research can enable LLMs trained with a finite attention window to work on text of infinite length and solve the problem of forgetting previous necessaries context. Hardware requirements are

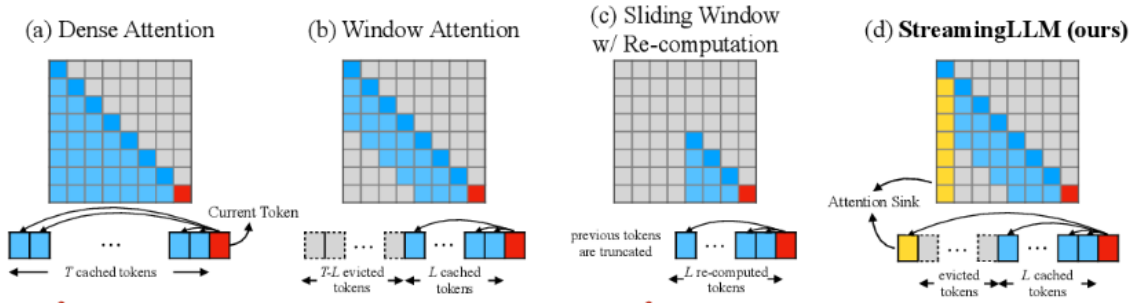


Figure 1.2: Visual representation of StreamingLLM

also considered as limitations . For performing the fine tuning of pretrained LLMs

model ,the powerful GPU and huge memory capacity is essential.Additionally ,for real-time prediction the speed for training the datasets should be more fast and generate accurate sentiment of text.So advance hardware infrastructure is an important for generating the more optimized engine.For our proposed sentiment analysis engine,it also need to reduce the cost of training time of pretrained LLMs model.That's why to reach the requirement for deploying our proposed sentiment analysis engine,a higher hardware requirement may face significant limitation for our study.

1.4 Research Objectives

This research aim is to develop an engine that can be able to predict the insights sentiment of transliterated Bengali text by using Large language Model.In this research ,it also shows the comparison between traditional machine learning approaches and Large Language Model methods for predicting sentiment of transliterated bengali text.In order to do that , we will be using daraz user reviews datasets to finetune the pretrained LLM model. The objectives of this research are:

- To understand the complex contextual meaning,sarcasm of transliterated bengali text by using LLM model.
- To develop an engine that can get any collection of comments as an input and give the sentiment analysis results as an output.
- Our proposed engine will be able to run in client machines and also in servers.
- To compare our proposed pretrained model with other traditional machine learning methods.
- This research will also overcome the limitations about high VRAM usage and forgetting necessary details which are related to Context Window Size.

Chapter 2

Literature Review

Sentiment analysis has recently become more focus of study. This is because of its wide variety of possible applications such as public opinion, customer feedback analysis social media monitoring.It is very difficult to extract the subjective information from textual data .This textual data can provide useful insights into the thoughts and opinions expressed by people. Researchers showed more interest in the field of analyzing the massive amounts of text published online.Traditionally, sentiment analysis depended on rule-based methods or traditional machine learning techniques .And it requires manual feature building processes. But these techniques faced many difficulties in dealing with the complexity and nuance of human language. However,the transition to Natural Language Processing (NLP) has transformed sentiment analysis more advanced. Notably, the use of Large Language Models (LLMs) such as GPT-3 and BERT has given new ways for further research . They can now identify complex verbal nuances and contextual information with incredible accuracy.Large Language Model has its extraordinary ability to learn complex language structures.That's why Large Language Models (LLMs) have changed the area of sentiment analysis. GPT-3, for example, stands out as a generative model capable of producing contextually relevant text.BERT improves at recognizing two-way context which helps it grasp sentence word relationships. These advancements allow the extraction of nuanced emotions from many textual sources .A turning point took place in the rise of this field. Because of these models' skills they can look at written events in a more complicated and context-aware way. This has caused a lot of changes to their settings.This big change could help in many ways.With these changes, sentiment analysis can now find the contextual feelings in a lot of different kinds of text.

2.1 Sentiment Analysis Techniques

In the early stages of sentiment analysis, rule-based methods established the criteria and contextual patterns to evaluate emotion polarity. C.J. Hutto and Eric Gilbert developed the VADER model specifically for sentiment analysis in social media content [3]. n their research ,the VADER has outperformed individual human raters and exhibited strong generalization across a wide range of conditions.This success shows that rule-based methods can be used to generate complex views from a lot of different types of textual data.These approaches depend basically on sentiment lexicons or dictionaries with pre-assigned sentiment scores for words. Based on these

lexicons, words were classified as positive, negative, or neutral .However, the overall sentiment of a document Linh Vu and Thanh Le suggested a lexicon-based sentiment analysis tool for social network data [4].Their research shows a different approach of combining SentiWordNet and LIU approaches with a heuristic data pre-processing strategy.This method showed better results when compared to previous lexicon-based strategies. However, still sentiment analysis struggled to understand sarcasm, conditional words, and objective statements using Lexicon-based approaches . These approaches depend basically on sentiment lexicons or dictionaries with pre-assigned sentiment scores for words. The early success of rule-based methods in sentiment analysis as well as their vital part in establishing the foundations for future improvements in the area. However, rule-based sentiment analysis approaches have some limitations. The inflexibility of predefined norms which may struggle to adapt to the dynamic and growing character of language is one difficulty. Also, these methods might have trouble with subtleties of language like humor or irony .In addition ,it might have face to figure out what people mean when they say things that aren't clear from the context.Sentiment analysis study is still going on to find ways to overcome these problems and make rule-based methods more flexible.The experts want to make rule-based models that can handle data from people who speak a lot of different languages. They want to make sentiment analysis models that perform better and more varied so that they can figure out the contextual meaning of people to say things in a lot of different situations. Both computer linguistics and natural language processing depend strongly on this.

Moreover significant developments were made with a shift towards traditional rule-based approaches . This development increased the possibilities of statistical and machine learning methodologies. The presence of labeled datasets played an important role in enabling this progress. That's how it became a fundamental transformation in sentiment analysis. This progress happened for these labeled datasets.It allowed models to overcome the limitations of rule based approach. Because it was possible to understand the patterns and correlations directly from the data . Thus,the research about sentiment analysis became increasingly flexible and data-driven .Researcher shows more interest in the field of Sentiment analysis that's why numerous machine learning classifiers are used.Models like Naive Bayes, Support Vector Machines (SVM), and Decision Trees have shown to be useful tools for sentiment prediction. Researchers have extensively studied their efficacy in many contexts.For example,Mullen and N. Collier introduced sentiment analysis methodology that used Support Vector Machines (SVMs) and integrated several input sources [5].The adaptability of contemporary machine learning techniques can be showcased by their research.Furthermore, future study has prioritized for enhancing and refining sentiment analysis methodologies.Ahmad,M. et al. presented the optimized Sentiment Analysis Framework (OSAF) which combines the Support Vector Machine (SVM) with a grid search technique [6].Despite acknowledging computational complexity issues, the study contributed to current attempts to improve the performance of sentiment analysis models.In more general terms, there has been impressive and continuing study into machine learning algorithms for sentiment analysis. Baid, P. et al.'s paper makes an important addition by diving into the analysis of movie reviews using several algorithms such as Naive Bayes, K-Nearest Neighbor and Random Forest[7].The experts were interested in seeing how well these algorithms worked when used in movie reviews.With an impressive success rate of 81.4% the study's results

clearly show that Naive Bayes is the best method for analyzing movie reviews.In this case, it shows how well machine learning systems can read text data about movies to find sentiment details.Research also applied machine learning techniques to measure sentiment of texts in a lot of different ways. Changes and improvements could always being made with algorithms to help them understand the complex feelings that are shown in text data . Machine learning can reveal how people interact and interpret about different topics.

In era of digital age,the power of Natural Language Processing (NLP) can open more ways for doing better in Sentiment analysis fields .These things are now linked in a way that change the nature of AI.Progress in sentiment analysis change the way of understanding complex contextual meaning of words and open up new opportunities in many areas. It also takes us to new places where we haven't been before.The accuracy of detecting emotions is improved by using NLP in sentiment analysis. It also enhanced the level of environmental knowledge that wasn't possible before it was added.And it makes the field of language analysis more complete. The introduction of pre-trained language models like BERT and GPT is an example of a groundbreaking development that shows how well they can capture complex environmental links within textual data.In 2018, Devlin et al. showed that these models can pick up on small differences in language[8]. This makes it even easier to tell the difference between moods and gives us new ways to understand words.Moreover, Zhang,X. have paved the way for an innovative approach that combines named entity recognition with sentiment analysis [9]. It introduced a new era of interdisciplinary fusion. This not only enhances sentiment analysis but also contributes to a broader understanding of textual data .This holistic approach showcases the multifaceted capabilities of NLP.It provides a comprehensive interpretation of human language dynamics.Cultural and linguistic diversity are the main challenges to the application of sentiment analysis in varied linguistic landscapes. Adaptable models that can understand slang, colloquialisms, and phrases that depend on the situation are being worked on by researchers to deal with these problems.Sentiment analysis in NLP is dynamic because it can change to the constantly changing language environment. This makes sure that it is relevant and accurate across a wide range of communication platforms.As time goes on and technology improves, mood analysis goes beyond just classifying feelings. It is being used more and more in NLP-based tasks .It also shows a more complete method for understanding language.Sentiment analysis and natural language processing (NLP) work well together to include multimedia elements like pictures and movies which shows how complicated human communication is becoming. By the using of NLP , sentiment analysis techniques has now better able to handle the complexities of language across different types of communication. This means that we can fully understand how people feel when they use different ways to describe their opinions.However,the modern world of sentiment analysis is made up of pre-trained language models and methods from different fields and the ability to respond to different language environments.As technology gets better, this methods are likely to help us understand human words better by taking into account context and culture.

In this ever-evolving landscape of Sentiment Analysis,Aspect-Based Sentiment Analysis (ABSA) is a crucial tool in understanding sentiment analysis tasks.This method can contribute a lot in the case of understanding the contextual meaning of written information. However,this process begins by extracting key aspects and then cat-

egorizing attitudes linked to each identified feature in a well-planned sequence.At the end of the day, these fine-grained findings help with both micro-level sentiment discernment and extracting the macro-level sentiment included in the whole text.Pavlopoulos (2014) stands out in ABSA techniques. He provides a clear path for future research by highlighting key features and thoroughly evaluating sentiments in texts describing specific items [10]. Pavlopoulos's methods can help us to understand emotions better .It also improves our knowledge of sentiment analysis by exploring how traits and emotions are connected and categorized.Moreover,a groundbreaking step in ABSA was taken by Jiang et al. (2019) with the introduction of the Massive Multi-Aspect Multi-Sentiment (MAMS) dataset [11].This dataset showcases sentences embodying two or more distinct components with different polarities of emotion. They presented advanced models such as CapsNet and CapsNet-BERT which outperformed existing methods.Exploring the architecture and training of these models is crucial for enhancing sentiment analysis capabilities.In the latest scholarly endeavor published in 2023, Ahmed,k. et al. delve into the augmentation of ABSA through the integration of contextual embeddings in aspect extraction [12]. Their methodological amalgamation of attention processes with adeptly trained language models manifests in the discernible improvement of subtlety identification. This latest development interacts well with continuing work to improve the accuracy and detail of ABSA, properly representing the ever-changing nature of sentiment analysis methods as they constantly develop.The insights from Ahmed,k. et al. not only contribute to the academic discourse but also increase the momentum of progress within the practical applications of sentiment analysis.The innovative sentiment categorization ensemble technique presented by Kim, k. et al. adds yet another unique layer to the expansive ABSA landscape [13].The ideas presented by Ahmed et al. not only add to the academic conversation, but they also speed up progress in using sentiment analysis in real life.But the new sentiment categorization ensemble method that Kim and Lee describe adds another unique layer to the huge ABSA environment.The study shows that guessing emotions might be more accurate if more than one machine learning model is used together. It shows how the connections between many models help each other and make estimates more accurate .We need to look at more than one model and build a good framework for sentiment analysis that can deal with the problems that come up when people have a lot of different feelings in different areas.This will enhance the practical usefulness of ABSA.The synthesized perspectives from Ahmed,k. et al. and Kim, k. et al. collectively underscore the perpetual evolution in the ABSA domain, shedding light on the diverse approaches and models that continually redefine the contours of sentiment analysis.

Nowadays Deep Learning Techniques create a historic milestone in sentiment analysis.Researchers usually use neural network architecture for getting complicated patterns. Many scholars have contributed to develop sentiment analysis by utilizing pretrained models such as ELMo and BERT on extensive datasets. These findings not only enhance our understanding of the field but also showcase cutting-edge results.It also provides a comprehensive perspective on the situation.Araci, D. proposed FinBERT, a BERT-based language model that can optimize financial sentiment analysis by requiring fewer samples with labels and fine-tuning on a domain-specific corpus [14]. It outperforms state-of-the-art machine learning approaches as well as producing better results with a smaller training set.But In some circumstances,

the model could produce incorrect predictions and the rest of the model must still learn complex relations with a limited amount of labeled data.However,the accuracy rate of FinBERT on a specific task or dataset is 86% .In addition,Zhang, T. et al. execute a methodical evaluation of five existing SA4SE tools and versions of four state-of-the-art pre-trained Transformer-based models [15]. They present the first fine-tuning of pre-trained Transformer-based models for the SA4SE job.However,this study demonstrates the utility of pre-trained Transformer models, such as BERT and XLNet, for enhancing sentiment analysis in the software engineering domain, indicating to a viable possibilities for future research and development in this area.

2.2 Sentiment analysis of Banglish Text

Banglish sentiment analysis is a field that is always changing. Scholars are exploring numerous approaches in the field of Banglish sentiment analysis where English and Bengali words are mixed. However , understanding the contextual meaning of this language creates unique challenges for researchers to investigate innovative methods. Hossain et al.'s (2022) prediction model which was based on the desire for smartphones in the Bangladesh market [16]. It was very good at identifying named entities. Using both machine learning and natural language processing (NLP) methods together made the model even better at understanding how to handle complex Banglish user input. Getting high scores of 95.51% on Amazon Comprehend Custom NER and 87.99% on Spacy Custom NER shows that the model is good at getting useful information from different types of language input .Alam et al. (2017) made a big step forward in Bangla sentiment analysis by creating a system built on Convolutional Neural Networks (CNNs) [17].The model had a classification accuracy of 99.87% which was higher than other sentiment classifiers. This shows that deep learning has the ability to pick up on subtleties of emotion in Bangla texts. Carefully selecting 850 Bangla comments, labeled as either positive or negative, showed that the model was good at reading emotions in different kinds of language .Moreover,The work that Tripto and Eunus did to look into sentiment analysis and expression recognition from Bangla YouTube comments was very helpful to the field [18].Using deep learning methods, they got very good results which was 65.97% for three-label sentiment classification and 54.24% for five-label sentiment classification. Texts specific to the subject and language were found to be better worked by their method than Romanized Bangla. This showed how flexible and useful their methods are for picking up on subtleties of emotion (Tripto, N., & Eunus, M.).As experts keep looking into the complexities of Banglish mood analysis, new ways to make it better keep showing up. Putting together rule-based systems with advanced deep learning models is one possible direction. Certain subtleties of Banglish language can be picked up on by rule-based systems and context knowledge for more accurate mood analysis is added by deep learning models. Adding cultural and social factors that are specific to Banglish conversation can also make mood analysis models better. To improve models for real-world use, it's important to know how regional differences affect how people share their feelings in Banglish. For the full growth of Banglish mood analysis methods, people from different language, cultural, and technical backgrounds must work together.For businesses that want to connect with people who speak this mixed language, the improvements in Banglish mood research are very important. Being able to correctly read the tone of Banglish

user comments on everything from social media sites to e-commerce sites makes exchanges more personalized and quick.Evolving Banglish as a dynamic language mix also requires constant adaptation and new ideas in mood analysis methods. Researchers may look into how Banglish speakers show their feelings over time, taking into account how language trends and phrases change over time. Longitudinal studies could tell us a lot about how opinion changes over time in this language setting. Banglish mood analysis has effects on society that are worth looking into, beyond its uses in education and business. Understanding how cultures are changing, how language choices are changing, and how conversation patterns are changing can be achieved by figuring out how people feel when they speak Banglish. This sociolinguistic perspective can contribute to a broader understanding of language dynamics in multicultural and multilingual societies. The dynamics of language in multilingual and multicultural cultures may be better understood from this sociolinguistic vantage point. The ongoing research into Banglish sentiment analysis is a final indication of a commitment to comprehend this peculiar amalgamation of languages. A scholarly contribution is additionally made by these endeavors, which could potentially be advantageous for industries aiming to navigate the dynamic and complex linguistic landscape of Banglish communication. Improved sentiment analysis models and a more nuanced understanding of the emotions of Banglish users comprise potential results of increased academic investment in this field. There is a lot of room for growth and development at the crossroads of language, culture, and technology in Banglish sentiment analysis.

2.3 Role of Language Models in Sentiment Analysis

Sentiment analysis has experienced substantial growth due to the utilization of language models, particularly Large Language Models (LLMs). An evolutionary shift toward more sophisticated models ensued as early machine learning algorithms were unable to capture the profound complexity of language. Significant advancements in natural language processing are exemplified by the progress made toward LLMs with GPT-3 and BERT. As a generative LLM, GPT-3 demonstrates the capacity to produce informative and accurate content. BERT recognizes intricate relationships between words in a phrase with remarkable proficiency due to its bidirectional comprehension. The myriad challenges that sentiment analysis tasks entail are addressed by these models owing to their adaptability and capacity to decipher intricate linguistic structures. Big steps forward in mood analysis and the use of language models have been made by a number of experts. The stage for future progress in this area has been set by their work.Research on the feasibility and effectiveness of utilizing Large Language Models (LLMs) for sentiment analysis tasks was conducted by academics. A comprehensive examination of current approaches utilized in the detection of LLM-generated texts is offered by the survey conducted by Tang, Chuang, and Hu [19].The importance of meticulous evaluation procedures and the danger posed by open-source LLMs is emphasized by it. The application of watermarks and classification models and categories detection methods as white-box or black-box is discussed in the article. Despite these developments, significant obstacles continue to exist in the detection of LLM-generated text, as observed by the

report.(2023). In order to address issues such as out-of-distribution generalization, optimal task planning, and AI self-improvement, Ge et al. [20]. incorporate novel techniques into the learning schema of LLMs by including RLTF and nonlinear task planning.
Moreover, with regard to financial sentiment analysis, early methodologies placed emphasis on refining pre-trained models through the utilization of financial sentiment analysis datasets. Nevertheless, these approaches encountered challenges when it came to identifying intricate financial news, specifically that which contained numerical data or lacked contextual information.NLP has witnessed the emergence of Large Language Models (LLMs), which have demonstrated robust capabilities in in-context learning and chain-of-thought analysis.Due to a misalignment between the training purpose of LLMs designed for the financial sector, such as BloombergGPT and FinGPT, and the objective of financial sentiment analysis, these models encounter difficulties in producing the anticipated sentiment labels.Zhang, B., et al. proposed a method for enhancing the precision and accuracy of financial sentiment analysis by combining instruction modification and regaining augmentation[21]. The limitations of traditional NLP models and the challenges unique to financial sentiment analysis were addressed by the authors.In the era of large language models, numerous researchers present their individual contributions.The article "Llama 2: Open Foundation and Fine-Tuned Chat Models" by Hugo Touvron et al. details the development and release of Llama 2, a collection of large language models (LLMs) optimized for dialogue use cases that were pretrained and fine-tuned.[19]The fine-tuning approach and safety enhancements are provided in this paper, enabling the community to further develop their contributions and actively participate in the responsible evolution of LLMs. The LLAMA2 model may be implemented in their own use cases by a multitude of researchers, thanks to this enormous contribution by Pavlyuchenko et al. present a novel approach in their paper for optimizing the Llama 2 GPT model for the processing of multimodal financial news by employing the PEFT/LoRA method[22]. The performance of the refined Llama 2 model in various tasks is intended to be examined by their research. Literature will be evaluated through the lens of financial markets and named entities with attitudes will be identified and extracted.

Chapter 3

Workplan

Our primary goal is to make a full optimized engine that can be able to get any collection of datasets as an input and give sentiment results as an output.For that,we build three modules named “Input Module”,”Sentiment Analysis Engine module” and ”Output Module” .These module follow the modular approach.In our sentiment analysis module ,we implement the pre-trained LLAMA-2 model which can be able to fine-tuned for generating the sentiment of transliterated bengali text.However,we collect a diverse amounts of datasets from a popular website in Bangladesh named Daraz Bd.In this datasets ,it contains large amounts of transliterated bengali text with a sentiment labeling of three different classifications named “Positive”,”Negative” and “Neutral”.

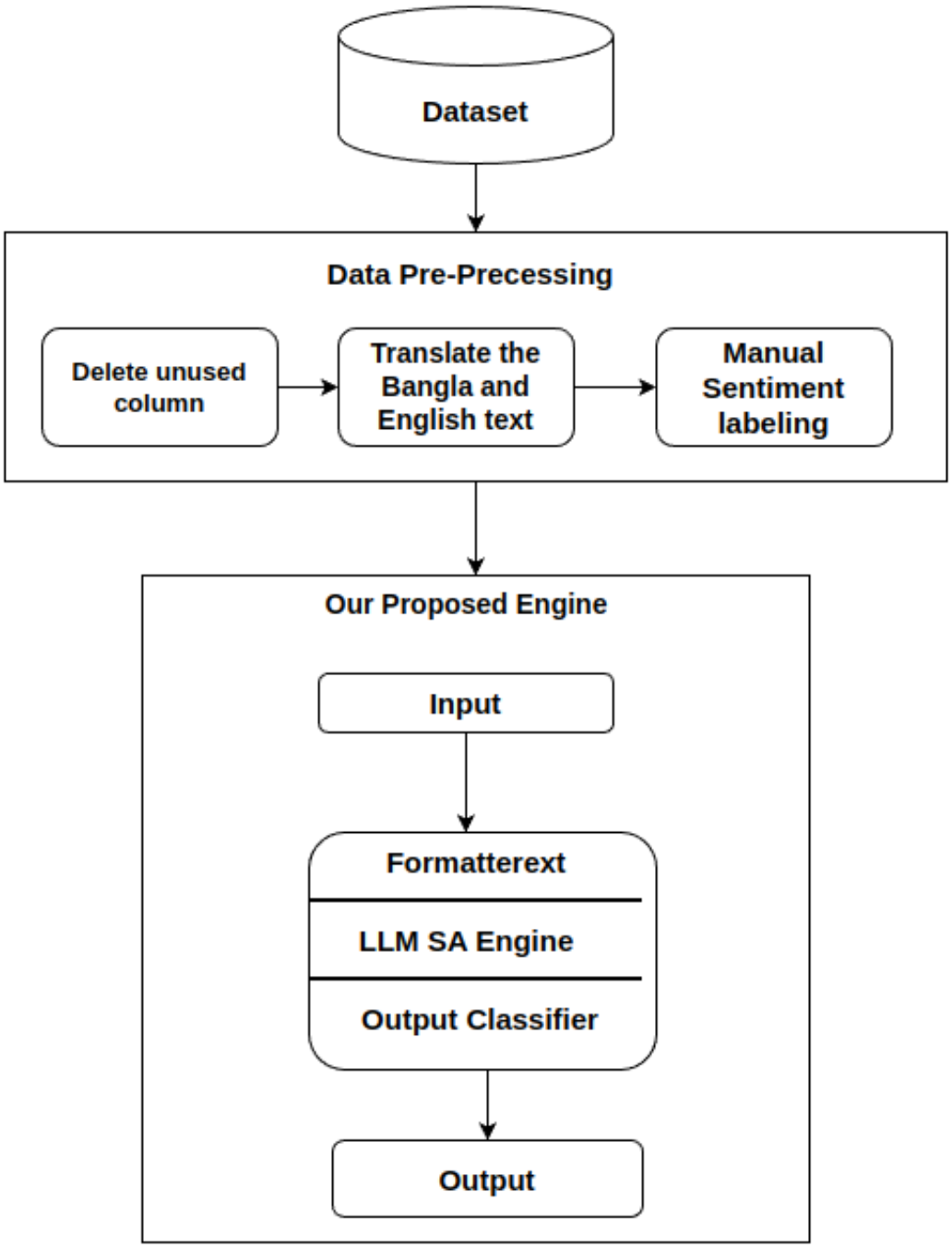


Figure 3.1: Overview of our Work Plan

Chapter 4

Description of the Datasets

4.1 Data Collection

Getting data is a very important part of building a strong dataset for real-time sentiment analysis in transliterated Bengali text. This research examines sentiment analysis using pre-trained language models. To do this successfully, you will need a dataset that contains both diverse and rich context. The chosen source for data collection is the prominent e-commerce platform, Daraz, operating in Bangladesh. The platform, Daraz, is targeted by us in order to capture customer sentiments and opinions.So that We collect the user's review from the different categories of products.However, the study focus on Daraz website as a data source because Daraz is the most popular ecommerce platform .That's why it contains a huge amount of users opinion from various products.

Daraz Sku	Price	rating	review_content
101174268_BD-1015184111	177	5	good product nice deliver
101174268_BD-1015184111	177	5	better
101174268_BD-1015184111	177	5	good
101174268_BD-1015184111	177	5	good
101174268_BD-1015184111	177	5	Good Product
101174268_BD-1015184111	177	3	Motamuti
101174268_BD-1015184111	177	5	Best product
101174268_BD-1015184111	177	5	This product was original and delivary process was awesome
101174268_BD-1015184111	177	5	মোটামুটি
101174268_BD-1015184111	177	5	প্রোডাক্ট ভালো ছিলো। সন্তুষ্ট আমি। ধন্যবাদ অনে কম দামে পেয়েছি।
101174268_BD-1015184111	177	5	Nice product in a nice packet.. Happy
101174268_BD-1015184111	177	5	good..thanks
101174268_BD-1015184111	177	5	ager products...dekhte besh puranoo
101174268_BD-1015184111	177	5	I am happy with buying this product. Thank you Daraz
101174268_BD-1015184111	177	5	এই প্রথম কিনলাম। কিছুদিন থেকে বাচ্চাদের মাখাচ্ছি। সব থেকে ভালো লেগেছে মুখ ল
101174268_BD-1015184111	177	5	nice smell
101174268_BD-1015184111	177	5	Good
101174268_BD-1015184111	177	5	শিশুর সঠিক ভাবে বেড়ে উঠার জন্য প্রয়োজন সুস্থ পরিবেশ, সুষম খাদ্য, এবং সঠিক
101174268_BD-1015184111	177	5	good, but smell ta arow sandor howa dorkar chilo baby product his
101174268_BD-1015184111	177	5	nice product.
101174268_BD-1015184111	177	5	আসল এবং ডেট ২০২৩ পর্যন্ত\n
101174268_BD-1015184111	177	5	My little bro is happy, smell is good enough \nthanks to daraz & Ma
101174268_BD-1015184111	177	5	Satisfied Quality is good
101174268_BD-1015184111	177	5	Good

Figure 4.1: Raw Datasets from Daraz

4.2 Data Preprocessing

This datasets contain about 17,335 user's opinions of 6 different categories named Rice,smartphones,earphones,baby lotion,shirt and moneybags.This datasets also contain the four columns named product code, price,rating and review_content.We started the data pre-processing part by dropping columns named product code, price and ratings.Because the data of those columns were not relevant to our research purposes.However,these datasets contain the mixture opinions of user-generated contents in three different languages such as English,Bangla and Bangla text written in English alphabets. As our research purpose is to deal with transliteration of Bengali text ,we keep only those reviews which contain the Banglish (Bangla language but written in English alphabets) from the review column. Regarding this purpose we translate manually from raw bangla text to Banglish (bangla language but written in English alphabets).

	A
1	**review_content**
2	valo chilo sobay nite paren
3	sera...puray..kop..mama
4	valoi sobay chalu niye nen..
5	valoi mone hoilo amar kache
6	Motamuti
7	overall valoi..josss
8	valo product....
9	product ta onek valo..😊😊
10	kidni beche product kinlam
11	product valoi chilo overall bolte gele
12	valo bolte thik e ache dam onujay
13	ager products...dekhte besh puranoo
14	onek josss...puray chokkham
15	onek kom dame pelam kintu amke manaitese na.ejonno mon kharap
16	valo sobay nissondehe niye felen...
17	dam kom holeo quality valo na
18	good, but smell ta arow sandor howa dorkar chilo baby product hisebe, parachute narikel tel er smell ase etai vallagena, otherwise good..😱
19	josss product.. keo na nile miss korbe.
20	beshi kharap na
21	My little bro is happy, smell is good enough .thanks to daraz & Marico Bangladesh 😉

Figure 4.2: Transliterated Bengali text

4.3 Sentiment Labeling

In the case of labeling parts I add another column named sentiment label.I want to train our pre-trained LLMs perfectly labeling dataset.In the case of vader sentiment analysis method for labeling ,it didn't recognize the contextual meaning of Banglish sentiment properly.That's why we chose manually labeling.The another reason for manually labeling is to overcome the limitations contextual meaning,sarcasm and cultural nuances.For example, It is very difficult to recognize the contextual meaning like "dam er dik vabte gele puray agun" -where llm may generate this sentiment of text is negative but the contextual meaning of this text is positive.However,we point out from the datasets that there are several more text contain like "puray makkhon","hat e peye pura matha nosto hoye gelo".This types of text is hard to understand the contextual meaning.So for overcoming this limitation ,in the sentiment label column we manually labeled the text in three classification named

positive,negative and neutral .To train our LLMs model for understanding the contextual meaning we add some more comments in review column with more complex text like ” matha nosto hoye gelo”,”kidni beche dia laglo”.

A	B
Sentiment Label	review_content
1	valo chilo sobay nite paren
1	sera...puray..kop..mama
1	valoi sobay chalu niye nen..
1	valoi mone hoilo amar kache
0	Motamuti
1	overall valoi..josss
1	**puray makkhon**
1	matha nosto product. ☺ ☺
-1	**kidni beche product kinlam**
1	product valoi chilo overall bolte gele
1	valo bolte thik e ache dam onujay
-1	ager products...dekhte besh puranoo
1	onek josss...puray chokkham
0	onek kom dame pelam kintu amke manaitese na.ejonno mon kharap
1	valo sobay nissondehe niye felen...
-1	dam kom holeo quality valo na
-1	good, but smell ta arow sandor howa dorkar chilo baby product hisebe, parachute narikel tel er smell ase etai vallagena, otherwise good.. 🤗
1	**hat e peye pura matha nosto hoye gelo**
1	beshi kharap na
1	valo chilo product ta
1	**puray agun mamur beta**
0	sera dam e jinista pailam
1	best lotion ei price e peye gelam..actually joss..sobay parle nite paren
0	aro valo aso korsilam

Figure 4.3: Manual Sentiment Labeling of Transliterated Bengali text

Chapter 5

Description of the model

5.1 Explanation about the Traditional Machine Learning Methods VS Our Proposed Model

By using large language models (LLMs), sentiment analysis has become more advanced to learn the complex nuances and contextual meaning.Traditional methods still hold their ground but the capabilities of LLMs offer a fresh perspective and unlock new possibilities. The initial step in our methodology involves the selection of a pre-trained LLM with LLAMA-2.This research chose LLAMA 2 model due to its state-of-the-art capabilities in natural language understanding. LLAMA-2 has been pre-trained on a vast and diverse amount which provides a solid foundation for further fine-tuning for sentiment analysis. It's important to acknowledge that each

Table 5.1: Comparison between Traditional and LLM-based Approaches

Comparison Criteria	**Traditional Approach**	**Our Proposed Model**
Data Dependency	Heavily depends on manually crafted lexicons and rule-based algorithms. Limited adaptability and domain-specific.	Leverages vast pre-trained data, learns statistical patterns. More adaptable and generalizable.
Contextual Understanding	Often struggles with sarcasm, irony, and complex sentence structures. Lacks deeper context awareness.	Excels at capturing context and relationships between words. Stronger understanding of nuances and implicit meanings.
Adaptability and Generalization	Brittle and domain-specific. Requires re-engineering and lexicon adjustments for different tasks.	Highly adaptable and generalizable. Fine-tuning for specific domains further enhances accuracy.
Scalability and Efficiency	Computationally expensive for large datasets. Lexicon creation and feature engineering time-consuming.	Highly scalable and efficient. Handles large datasets well with reduced processing time.
Interpretability and Explainability	Often lacks transparency in decision-making processes. Black-box algorithms and handcrafted rules obscure reasoning.	Can offer some level of interpretability (e.g., attention mechanisms). Still evolving in this area.

approach has its own strengths and weaknesses. Traditional methods can be less adaptable but it can be more efficient for specific tasks especially when the domain is well-defined and the lexicon is carefully crafted. Meanwhile, LLMs require larger datasets and computational resources but offer greater flexibility.In addition ,it can handle more complex and nuanced language.By understanding the core differences between these two approaches we can leverage the strengths of each to unlock the full potential of sentiment analysis .That's why it can get complex meaning from any textual data.Large Language Models (LLMs) can provide the most valuable tools in natural language processing (NLP) tasks that can leverage their pretrained capabilities in transliteration and tokenization to enhance various applications.

5.2 Functionality of Our Proposed Model

5.2.1 Transliteration Proficiency

Cross-Lingual Adaptability: Our proposed model are pretrained on massive multilingual datasets.Because of their transliteration ability , It can easily adapt in different script and language.However ,this pretrained model can be more beneficial for getting textual information of complex non-latin scripts.
Sound Understanding of Phonetics: The pretrained transliteration capabilities of this model can easily identify the deep complexity of phonetic similarities.This allows the model to convert characters from one script to another.In addition it can capture the complex patterns of words.That's why it can ensure the accurate representation of sentiments expressed across diverse linguistic domains.

5.2.2 Tokenization Expertise

Context-Aware Tokenization: LLMs is able to tokenize the text on its own. Because of this tokenization it can break down sentence into meaningful units. However this pretrained model can be troublesome in getting accurate results in context-aware tokenization sometimes.That's why finetuning this model is more beneficial in sentiment analysis.And for understanding the complex emotions.That's why it can lead to a more accurate analysis of sentiment in training phrases.
Emotion-Specific Tokenization: In the case of traditional machine learning tasks, it faces difficulties to understand the words that are strongly associated with specific emotions without a proper emotion depth map. Because of LLMs pretrained tokenization capabilities it can recognize emotion-specific words. This approach can help to understand the complex sentiment around a particular topic.In addition to that, it allows the model to identify and capture subtle emotional nuances in the text to better identify neutral sentiments.

5.2.3 Adaptability and Generalization

Transfer Learning Advantage: Transfer learning advantage allows the model to generalize well across different languages and contexts by leveraging patterns presented in similar language.In addition to that, this can help it be more adaptable to a wide array of real-world scenarios.

5.2.4 Backpropagation and Neural Network Optimization

The iterative adjustments for the model begins with backpropagation which is a fundamental technique in training neural networks. This process typically involves calculating the gradients of the model's error with respect to its parameters. These gradients guide the adjustments to the weights and biases in a way that minimizes the error.In addition, it can optimize the model for improved performance.
Our specific focus during backpropagation was on tailoring the model for sentiment analysis. This required the model to learn and emphasize features such as correct POS Tagging, propper phonetic transliteration and patterns of density in the data that are indicative of sentiment. These parameters helped decide the right moment for the model considered being ready for sentiment analysis. By labeling the dataset the model becomes more improved at recognizing and capturing sentiment-related nuances within the text.

5.2.5 Hyperparameter Tuning for Balance

- Striking the Right Balance:

  Hyperparameters are the external configuration settings of the model that play a crucial role in determining its behavior. Hyperparameter tuning involves to find the optimal values for parameters such as learning rates, regularization strengths, and batch sizes. Because of running the process multiple times it gave us an equilibrium point for striking the right balance to prevent common issues such as overfitting or underfitting.

- Preventing Overfitting:

  Overfitting occurs when the model can learn the training data too well especially in including its noise, leading to poor generalization to new, unseen data.By adjusting Hyperparameter tuning in Bangla comments that can build the ability to prevent overfitting .In addition, it also ensure the model that capture essential patterns without memorizing noise in the training data.

5.3 Data Splitting

Splitting the dataset into training and testing sets is a fundamental step in our model. It gives us the facility for model training, evaluation, and improvement.Moreover it also ensures that the model is robust and capable of making accurate predictions on new, unseen data.However ,for this research ,we split out data in four times.Below we give the details description of those splits.

5.3.1 0/100% Split (No Training Data for LLM)

This scenario serves as a baseline.The LLM is not exposed to any sentiment-labeled training data. Testing the model under this condition helps the ability to generalize the unseen sentiment expressions.We expect as a lower accuracy and F1-score.However,it provides valuable insights into the model's raw capability without any fine-tuning.

5.3.2 30/70% Split (Limited Training Data)

In this part we use 30% of the data used for training this split evaluates the model's performance. It exposes a limited amount of labeled sentiment data. The aim is to observe how well the LLM generalizes with a constrained training set.

5.3.3 50/50% Split (Limited Training Data)

This represents a balanced way where half of the dataset is used for training and the other half for testing. This split aims to ensure that the model is exposed to sufficient labeled data to learn sentiment patterns .

5.3.4 80/20% Split (Limited Training Data)

This part we use 80% of the data allocated for training.This configuration helps the model's performance when it exposed to a substantial amount of labeled sentiment data. The expectation is higher accuracy and F1-score.Because the model has more information to refine its understanding of sentiment nuances.

Chapter 6

LLM Configuration

Large Language Models (LLMs) have revolutionized natural language processing tasks, showing their remarkable capacity to understand and generate human-like text. Fulfilling optimal conduct in these models often involves fine-tuning, a process that tailors the pre-trained model to specific tasks or domains. Autotrain, an advanced automated training technique, has appeared as a powerful tool in the fine-tuning process, streamlining and enhancing the performance of LLMs.

6.1 Fine-tuning in LLMs

Large language models, such as OpenAI's GPT-3, are pre-trained on huge amounts of various data to grasp the complications of language. However, to denote specific tasks or domains, fine-tuning becomes important. Traditional fine-tuning methods involve manually adjusting meta parameters, a labor-intensive process that may not yield optimal solutions.

6.2 Auto Train: A Game-Changing Approach

Autotrain introduces automation to the fine-tuning process, seriously reducing the burden on practitioners. This method imposes advanced algorithms to frequently adjust metaparameters, learning rates, and other complex aspects, allowing the model to adapt manually to the specifics of the target task. Autotrain is especially effective in scenarios where manual fine-tuning may lead to suboptimal outcomes due to the complexity of the model and the huge parameter space.

6.2.1 Benefits of Autotrain in LLM Fine-Tuning:

- **Efficiency:** Autotrain pushes the fine-tuning process by automating parameter adjustments, opening up faster convergence and reducing the need for extensive manual intervention.
- **Optimization:** Autotrain looks at the parameter space more carefully, seeking optimal configurations that might be hard to identify through manual fine-tuning alone. This leads to improved model performance on specific tasks.

- **Generalization:** By automating the fine-tuning process, Autotrain aids in creating models that generalize well across diverse datasets, enhancing the robustness of LLMs in various applications.

- **Scalability:** Autotrain is able to handel, making it suitable for large datasets and physically intensive tasks. This adaptability ensures that fine-tuning remains feasible for a large range of applications, from small-scale projects to enterprise-level solutions.

6.2.2 Challenges and consideration of Auto-train

Autotrain is essential to recognize potential challenges. The automated nature of the technique may lead to overfitting issue. Operators must establish a balance between automation and human interaction to ensure the model's adaptability.

6.2.3 Advantages of Auto-train

Autotrain stands out as a attractive approach for fine-tuning Large Language Models (LLMs) in sentiment analysis. Here's a discussion on the advantages of Autotrain in this specific context:

- **Adaptability to Complex Sentiment Patterns:**
  Sentiment analysis often needs models to capture subtle and complex patterns in language, ranging from subtle positive or negative complexes to sarcasm. Autotrain's adaptive adjustments allow the LLM to adapt its parameters more effectively, enabling it to grasp intricate sentiment nuances that might be complex for statically quantized models.

- **Efficient Exploration of Hyperparameter Space:**
  Autotrain excels at exploring the metaparameter realm proficiently. In sentiment analysis, finding the right equilibrium between model complexity and simplicity is critical. Autotrain's capacity to iteratively examine various metaparameter setups aids in pinpointing optimal settings for sentiment-associated tasks, improving overall model performance.

- **Automated Optimization for Sentiment-Specific Features:**
  Sentimental analysis often needs the model to focus on specific linguistic features relevant to emotions and attitudes. Autotrain, by automating the optimization process, can better fine-tune the LLM to emphasize sentiment-specific features, leading to better accuracy and effectiveness in sentiment classification.

- **Robust Generalization Across Diverse Sentiment Datasets:**
  Sentiment analysis models should perform well across diverse datasets that may contain varying language styles, expressions, or cultural influences. Autotrain's overreaching exploration of parameter space enhances the model's ability to generalize, making it more robust when applied to different sentiment analysis scenarios.

- **Time and Resource Efficiency:**
  Autotrains are a time-efficient way, as they reduce the need for human improvement during the fine-tuning process. Autotrain's automatic technique ensures a prompt reaction to emerging sentiment patterns without requiring laborious human tuning—a critical capability in sentiment research, where quick adaptation to shifting attitudes in social media or consumer evaluations is essential.

- **Scalability for Large-Scale Sentiment Analysis Applications:**
  Autotrain's capability is particularly useful for large-scale sentiment analysis applications. As the volume of sentiment-related data grows, Autotrain can effectively manage the fine-tuning process, ensuring that sentiment analysis models remain effective and accurate on extensive datasets without compromising performance.

While quantization and other tuning ways have their merits, Autotrain's adaptability, efficiency, and automated optimization make it particularly suitable for the dynamic and nuanced nature of sentiment analysis tasks. As the demand for sentiment analysis in various domains continues to rise, Autotrain presents itself as a valuable tool for unlocking the full potential of LLMs in capturing sentiment nuances accurately and efficiently.

6.3 Requirements

- Python 3.7 or higher
- TensorFlow 2.x
- Hugging Face Transformers library
- Autotrain library (ensureing the latest version is installed)

6.3.1 Token Size Considerations

Given the specified token size of 64, ensure that the average length of the input and output sequences combined does not exceed this limit. Tokenizing longer sequences might result in truncation or information loss.

6.4 LLAMA2 Model

Large Language Models (LLMs) learn to understand sentiment labels by capturing patterns in text through the vast amount of training data Here's how LLMs, such as the "meta/llama-2-7b-chat-hf" model may accomplish this task:

- **Semantic Understanding:** Llama 2 trained on diverse text sources that can develop a deep understanding of semantics, contextual relationships and language nuances. They learn to associate words, phrases, and context with different sentiment expressions.

- **Contextual Embeddings:** This LLama 2 model can generate contextual implants for each word in a sequence by considering the surrounding context. This enables them to capture intricate relationships between words which is crucial for understanding the sentiment-related patterns.

- **Pattern Recognition:** Our pre-trained Llama 2 has their own ability to recognize complex patterns connected with specific sentiment labels. These patterns may include the choice of words, syntactic framing, and contextual cues that signify positive, neutral, or negative sentiments.

- **Representation of Sentiment Labels:** Sentiment labels, such as [-1, 0, 1], are connected with corresponding sentiment expressions (e.g., negative, neutral, positive). LLMs learn to represent these labels in their internal representations, linking them to the patterns they have identified in the training data.

- **Training with Supervised Labels:** In the fine-tuning process for sentiment analysis tasks, LLMs are exposed to labeled datasets where each input text is associated with a sentiment label. The model adjusts its parameters to minimize the difference between its predicted sentiment scores and the ground truth labels, effectively learning to predict sentiment.

- **Transfer Learning:** The pre-training phase of LLMs involves learning general language patterns. The fine-tuning process refines the model's acknowledgement of sentiment-specific patterns by leveraging task-specific data. This transfer learning approach allows LLMs to adapt their knowledge gained from general language understanding to sentiment analysis tasks.

- **Incorporation of External Context:** LLMs may also contain external context from the training data, learning to understand sentiment in specific domains, industries, or cultural contexts. This capacity contributes to their ability to predict sentiment in diverse scenarios.

By leveraging these mechanisms, LLMs like "meta/llama-2-7b-chat-hf" can effectively learn and generalize sentiment patterns. This acquired knowledge enables them to make predictions about sentiment labels for unseen text in the future, making them valuable tools for sentiment analysis tasks across various domains.

Chapter 7

Implementation

The proposed engine is composed of three essential modules. (i) Input Module, (ii) Output Module, (iii) Sentiment Analyzer. Firstly, we implemented the Input Module in Python to accurately take the data from the standard input pipeline of the native os or a TCP socket. Our engine also offers a versatile sanitization handling mechanism by enabling callback input dispatcher. The Sentiment Analyzer Module is also developed in Python that serves as a wrapper around the 'transformer' library from Hugging Face. The analyzer is able to wrap any text generator that can talk to the other modules through STD I/O pipeline or TCP socket. Finally, the Output Module is also implemented in Python that deals with the dispatching of the sentiment analysis results. This module accommodates different use cases by supporting output through both the standard output pipeline and sockets. The interplay between these three modules ensures a flexible and efficient sentiment analysis engine, suitable for integration into diverse applications and systems.
This modular approach in the design of the engine sets a solid foundation for easy integration into browser-based add-ons or software plugins. The Input Module's adaptability allows it to easily interface with diverse sources that accommodate varied contexts of browser-based interactions. This flexibility ensures compatibility with different input streams commonly found in browser environments or add-ons targeted towards a particular piece of software. The Sentiment Analyzer Module's encapsulation of the transformer library serves as a robust analytical core that can be easily plugged into various software frameworks. Finally, the Output Module's support for different output channels makes it customizable in tailoring the presentation of sentiment analysis results for specific needs within the required interface. By encapsulating functionality in well-defined modules, the engine becomes highly portable which simplifies the process of embedding sentiment analysis capabilities wherever needed. Be it in browser add-ons or software plugins the adaptability for extending the reach of sentiment analysis to diverse applications and platforms becomes more feasible.
This architecture comprises three main modules:

- Input Module: Implemented in Python, this module accepts input either from the standard input pipeline or a socket. Designed to handle diverse input sources seamlessly, ensuring flexibility in data ingestion.

- Sentiment Analyzer Module: Developed as a Python wrapper around a transformer library, this module performs sentiment analysis on the input data.

Leverages state-of-the-art natural language processing techniques to extract sentiment information efficiently.

- Output Module: Also implemented in Python, this module dispatches the analyzed output in the required format. Supports output delivery through both standard output pipeline and sockets, catering to various use cases.

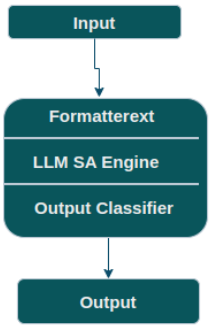


Figure 7.1: Overview of modular approach

7.1 Input Module

The Input Module's mechanism allows for the easy configuration of input sanitization processes,It ensures that the data entering the system meets specific requirements or standards. This configurable layer introduced via layered structure enhances the engine's resilience to diverse input scenarios.It is a crucial aspect for browser-based add-ons dealing with varied content sources.However,this module integration of both standard I/O and socket implementation enhances the engine's modularity, making it adaptable to various user interfaces.Below we give the algorithms of our input module.

```
function get_text_input():
    source = input("Enter 'stdin' for standard input or 'socket' for TCP socket").toLowerCase()
    if source is 'stdin':
        text_input = input("Enter the English sentence")
    else if source is 'socket':
        host = input("Enter the TCP server host")
        port = parseInt(input("Enter the TCP server port"))
        create a TCP socket s
        connect s to (host, port)
        text_input = receive data from s, decode it
    else:
        print("Invalid input source. Please enter 'stdin' or 'socket'.")
        text_input = None
    return text_input
function generate_sentiment_command(sentence, sentiment):
    command = "<BS>" + sentence + "</BS><SA>" + sentiment + "</SA>"
    return command
function main():
    sentence = get_text_input()
    if sentence is not None:
        sentiment_prediction = [-1, 0, 1]
        sentiment_command = generate_sentiment_command(sentence, sentiment_prediction)
        print("Dispatching request:", sentiment_command)
if __name__ is "__main__":
    call main()
```

Figure 7.2: Algorithm of Input Module

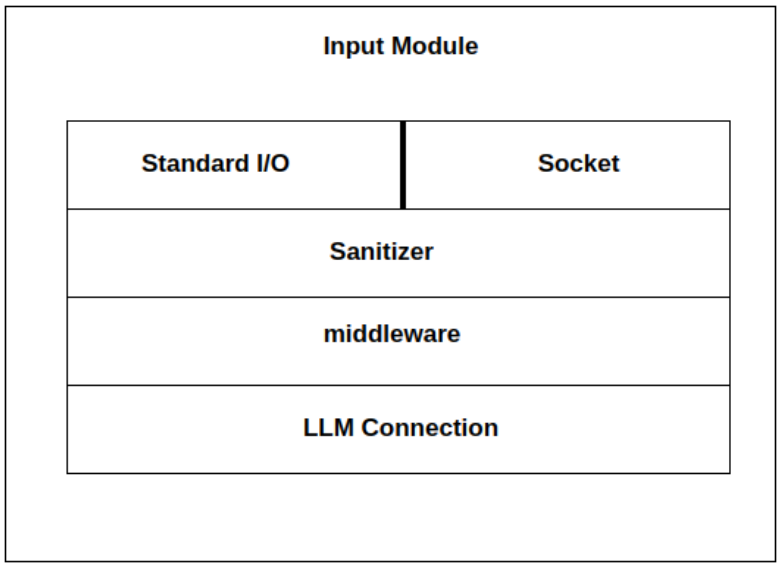


Figure 7.3: Overview of Input Module

Our proposed input module can be able to choose the input source (standard input or TCP socket). If a TCP socket is selected, it asks for the server's host and port. Then, it obtains the English sentence and generates a sentiment prediction command using a placeholder sentiment prediction. Finally, it dispatches the request by printing the formatted command in a specific format, demonstrating a basic framework for interacting with an LLM engine for sentiment analysis.This modular design allows the engine to seamlessly integrate with different interfaces, enabling users to interact through standard input or via network communication. Such versatility facilitates the integration of the sentiment analysis engine into diverse applications with distinct input preferences, promoting adaptability and ease of use.
This modular architecture of this sentiment analysis engine not only provides adaptability for integration but also introduces configurability in terms of input sanitization or output formatting. The Input Module's event based dispatching mechanism allows for the easy configuration of input sanitization processes, ensuring that the data entering the system meets specific requirements or standards. This configurable layer introduced via layered structure enhances the engine's resilience to diverse input scenarios, a crucial aspect for browser-based add-ons dealing with varied content sources.

7.2 Output Module

We create an output module in Python that takes input text, sends it to a TCP socket, receives sentiment predictions from a Language Model (LLM) engine, and then dispatches the response in the specified format. Here's an algorithms of implementation:

```
function send_to_socket(message, host='localhost', port=8888):
    create a socket s
    connect s to (host, port)
    send encoded message to s
    receive response from s, decode it
    return response

function generate_output(input_text):
    sentiment_prediction = send_to_socket(input_text)
    output = "<BS>" + input_text + "</BS><SP>" + sentiment_prediction + "</SP>"
    return output

if __name__ is "__main__":
    input_text = "This is an example sentence."
    output_result = generate_output(input_text)
    print("Output Result:", output_result)
```

Figure 7.4: Algorithm of Output Module

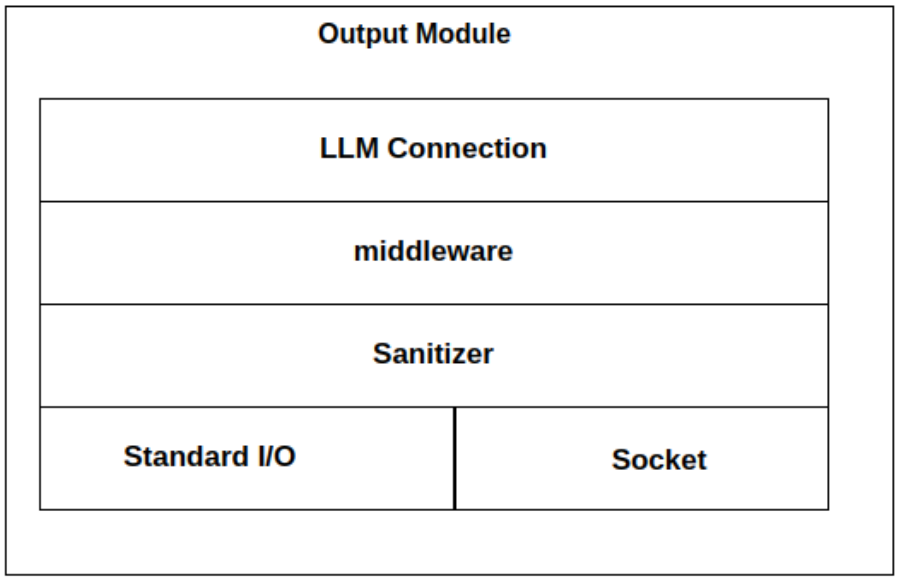


Figure 7.5: Overview of Output Module

The main functionality is encapsulated in the `generate_output` function. It takes an input text, utilizes the `send_to_socket` function to send this text to the LLM engine, and receives the sentiment prediction as a response. The output is then formatted according to the specified structure: `<BS>BANGLISH SENTENCE</BS><SP>[-1,0,1]</SP>`. The `<BS>` tag encapsulates the original input text, and the `<SP>` tag contains the sentiment prediction in the form of a list `[-1, 0, 1]`.
However, the output module plays a pivotal role in presenting results derived from the sentiment analysis module, specifically the fine-tuned LLAMA-2 model for Banglish text. It serves as the interface through which users or other systems can interact with the sentiment predictions generated by the analyzer. The module utilizes both standard output and a TCP socket to provide flexibility in how results are showcased. This design ensures a standardized and easily interpretable output. Whether displayed through the standard output pipeline or transmitted over a TCP socket, the output module acts as a versatile bridge, enabling seamless integration with

diverse applications and interfaces, contributing to a user-friendly and adaptable sentiment analysis solution.

7.3 Sentiment Analysis Module

Our proposed sentiment analysis module defines a system for generating output from a Language Model (LLM) engine and formatting it into a specific structure. The `send_to_socket` function establishes a TCP socket connection and sends a given message to a specified host and port. The response from the socket is then received and returned.However,here's an algorithms of implementation:

```
function sentiment_analyzer(bangla_sentence):
    # Load pre-trained tokenizer and model
    tokenizer = AutoTokenizer.from_pretrained("your_pretrained_model_name")
    model = AutoModelForSequenceClassification.from_pretrained("your_pretrained_model_name")

    # Preprocess input
    input_text = '<BS>' + bangla_sentence + '</BS>'
    input_ids = tokenizer(input_text, return_tensors="pt").input_ids

    # Make prediction
    disable gradient computation
    logits = model(input_ids).logits

    # Get sentiment prediction
    sentiment_prediction = argmax(logits, axis=1).item() - 1

    # Prepare response
    response = '<BS>' + bangla_sentence + '</BS><SP>' + '[' + sentiment_prediction + ']</SP>'

    return response

# Example usage
bangla_sentence = "Your Bangla sentence goes here."
result = sentiment_analyzer(bangla_sentence)
print(result)
```

Figure 7.6: Algorithm of Input Module

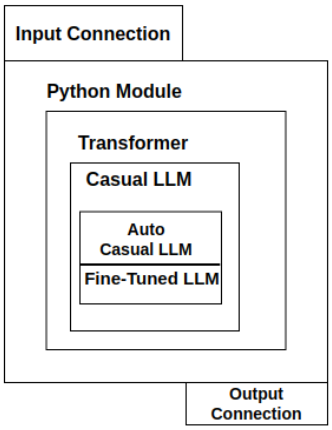


Figure 7.7: Overview of Sentiment Analysis Module

The sentiment analyzer engine is actually based on the pre-trained LLAMA-2 model for Banglish text and serves as a powerful tool for discerning sentiments in text data. Leveraging pre-trained knowledge and specific adaptations for Banglish, our

engine assesses the emotional tone of input sentences. Employing advanced natural language processing techniques, it assigns sentiment scores, typically represented as a list such as [-1, 0, 1]. This comprehensive sentiment analysis engine enables applications to understand and respond to user emotions in Banglish, enhancing user experience and facilitating tailored interactions in various domains such as social media monitoring, customer feedback analysis, and content personalization.
The modular design of the I/O system in the provided code enhances the overall modularity of the sentiment analysis engine, enabling an add-on level plug-and-play capability. By encapsulating the input processing, model prediction, and response generation within the `sentiment_analyzer` function, each component becomes a distinct module. This modular approach allows for easy integration of additional functionalities or replacement of specific modules without disrupting the entire system. For instance, one can seamlessly add new features, update the tokenizer, or swap the pre-trained model while maintaining compatibility with the existing code. This plug-and-play ability fosters a flexible and extensible architecture, facilitating the incorporation of future enhancements or alternative components without the need for extensive code modifications. Ultimately, this modularity contributes to the adaptability and scalability of the sentiment analysis engine.

Chapter 8

Result & Analysis

In the result and analysis chapter of the thesis, our exploration into leveraging pretrained language models for real-time sentiment analysis in transliterated Bengali text has yielded insightful findings.
However,After completely finishing our training and testing process we analyze our Result.Our main research purposes is to build a real-time LLM engine .For real-time LLM engine ,minimize the training time is very important factors.And the other factors is our proposed model must have the ability to understand about the complex contextual meaning of transliterated bengali text in a most accurate way.That's why,our main performance matrix for analyze our results are training time and accuracy rate.In this case we choose two paper from previous works which is actually related about transliterated bengali text .We applied our datasets in the models used by the two previous papers and checked the Accuracy Rate and Training Time. Below,we show the overall results:

Table 8.1: Model Accuracy rate

Model	**Accuracy (%)**	**Training Time (hours)**
Amazon Comprehend Custom NER	90.2	9.47
CNN Model	93.2	8.34
Our Proposed Model	83.47	40.02 minutes

The comparative analysis highlights that LLM outperforms its training time, exhibiting a superior time of 40.02 minutes. This suggests that leveraging pretrained language models in real-time sentiment analysis for transliterated Bengali text proves to be more effective than others models.

Table 8.2: Accuracy Rates of Data Splits

Data Split	**Accuracy Rate (%)**
0/100%	73.87
30/70%	76.5
50/50%	78.8
80/20%	84.37

Breaking down the accuracy rate further, the LLM's success can be attributed to its ability to capture intricate patterns and nuances in the case of data split

0/100%.However in this scenario the model was trained on the entire datasets and the accuracy rate is 73.87%.This could indicate that our LLM model can have the strong ability to capture the sentiment our datasets without any testing.However the accuracy improves to 76.5% when 30% of the data is reserved for testing. This suggests that the model is generalizing better at the time of evaluating a portion of unseen data.And if we set data split to 50/50 , the accuracy further increases to 78.8%.This indicates that the model can improve more from a equal training and testing split of data.Finally,the accuracy increases significantly to 84.37% when only 20% of the data is used for testing.

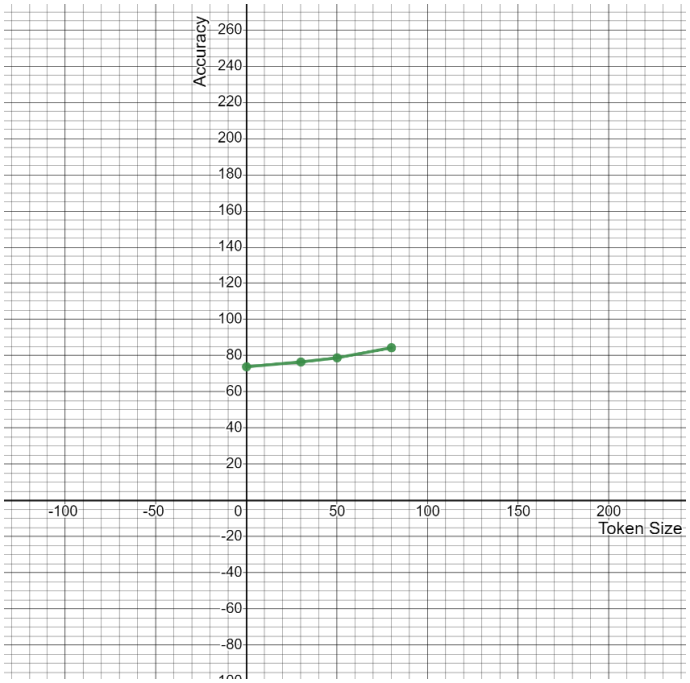


Figure 8.1: Graph of accuracy vs Tokensize

This is the above graph of accuracy vs Tokensize.According to the graph ,as we move from 0/100% to 80/20% ,there is a clear picture about the increasing trend in accuracy rate.This suggests that the model is improving as it is exposed to more diverse data during training and testing.

8.1 Calculating Slope:

To calculate the slope (m) of a line ,we use the tokenize data and accuracy rate from the table 8.2
The slope formula is

$$m = \frac{x_2 - x_1}{y_2 - y_1}$$

Let's use the points (0, 73.87) and (80, 84.37) from our table:

$$\begin{aligned} m &= \frac{80 - 0}{84.37 - 73.87} \\ &= \frac{10.5}{80} \\ &= \frac{21}{100} \\ &= 0.13125 \end{aligned}$$

This slope represents the rate of change in accuracy with respect to the change in the proportion of training data.This means that for every 1% increase in the proportion of training data, the accuracy is expected to increase by about 0.13125%.However the positive value of the slope suggests a positive correlation between the proportion of training data and accuracy.The result got from the slop calculation has influence all the members of this thesis to believe that a pre-trained LLM particularly the ones we tested in our research is already at a viable position for sentiment analysis in real-time for transliterated language.

Chapter 9

Conclusion & Future Work

The goal of this study was to be helped in understanding feelings better in Bengali texts by pretrained LLMs that have already been trained on big datasets. The challenges we faced for the language's unique alphabet and complex pattern are overcome by the study's entire focus on transliterated Bengali text. Thisdays Bengali words are actively translated by real-time sentiment analysis language models to understand emotions, with applications ranging from social media monitoring to customer feedback analysis. The main reason for using Large Language Models (LLMs) in this thesis is their cost-effectiveness and exceptional user-friendly which is a effective alternative to traditional sentiment analysis approaches.LLMs, which have undergone pre-training on vast datasets are emerging as a possible answer to the complex linguistic structure of Bengali, a language that is written using English alphabets.The conventional methods, such as Support Vector Machines (SVM) and K-Nearest Neighbors (KNN), face substantial challenges in capturing the subtleties of sentiment expressed in Bengali text. The particular syntax and linguistic complexities of transliterated Bengali text and the limitations of these traditional approaches make the use of LLMs a natural advancement in the quest for improved accuracy in real-time emotion analysis.
However, applying LLMs raises important issues that need more research.One of the biggest problems is that there is a lot of demand for Video Random Access Memory (VRAM) especially for big models like LLM. This could be a problem especially on devices that can't handle a lot of images which would make it harder for a lot of people to use mood analysis apps on a lot of different platforms. Also, LLMs' set context window widths might not be big enough to catch the whole context of a text which makes it harder for them to understand the complex feelings that are expressed in words or papers. Pretrained models offer the greatest importance of the contextual understanding, despite these difficulties, as recognized by the thesis. Having this knowledge is essential when dealing with the complex aspects of sentiment analysis especially in languages such as Bengali that are written in English alphabets.The study's goal is to improve methods for mood analysis by looking at both factors that affect the analysis and limitations. We want to find a way to deal with the difficult language patterns that people use in real life that works accurately and effectively. The future contributions outlined in this thesis underscore the commitment to advancing natural language processing, specifically in the context of sentiment analysis for transliterated Bengali text. Fine-tuning LLMs for Bengali sentiment analysis is identified as a crucial endeavor, aiming to capture the

subtleties inherent in Bengali language expression. Additionally, the emphasis on optimizing the language model for minimal hardware requirements aligns with the goal of making sophisticated language processing accessible to a broader audience. Moreover, one of the most important contributions will be to make the pretrained language model engine more flexible and economical in its use of resources. The goal of guaranteeing that the model works perfectly on minimal hardware setups, allowing for better accessibility to a larger audience is the development process. Especially important in various computing systems, where resource limits might make it difficult to implement complex language models, this is a particularly important point to keep in mind. So that everyone may utilize pre-trained language models for sentiment analysis, the research intends to make optimization easy. A wider audience will be reached in this way. The intentions of the thesis are excited to be put into action by creating a comprehensive sentiment analysis system that can extract perspectives from user comments across many platforms. In this day and age, when social media, online shopping, and online reviews are the main ways people talk to each other, companies, politicians, and researchers need to know how users feel. The system that is being suggested has the objective of processing and analyzing user-generated material from a variety of sources, including social media, e-commerce platforms, and restaurant reviews, in an effective manner. Real-time insights into user sentiments are intended to be delivered by this solution by leveraging the power of the fine-tuned and optimized pretrained language model. Informed choices will be enabled to be made by enterprises and user experiences will be improved to a greater extent.

In conclusion, this thesis establishes a solid foundation for the integration of pre-trained language models in real-time sentiment analysis for transliterated Bengali text. The outlined future contributions reflect a dedication to improving the capabilities and accessibility of language models, with a focus on cultural and linguistic relevance. The commitment to practical applications, exemplified by the proposed sentiment analysis system, underlines the research's relevance in addressing the dynamic challenges of the contemporary digital landscape.As the digital era continues to evolve, the significance of understanding and interpreting user sentiments becomes increasingly pivotal. This thesis's forward-looking approach not only contributes to the academic discourse but also strives to make meaningful advancements with practical implications in real-world scenarios. The commitment to innovation and the application of research findings positions this work as a valuable contribution to the ongoing narrative in natural language processing and sentiment analysis.

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